## Tools and Learning Resources

You would be forgiven for believing that a major part of the internet was actually composed of Python tutorials. Even so, Mark Pilgim's "Dive into Python" and "Dive into Python 3" books are great introductory texts. Another great book is "Think Python: How to Think Like a Computer Scientist". It too is available for free online at *www.greenteapress.com*

The folks at *www.interactivepython.org* have also converted the above book, among others, into an interactive online version that lets you run code samples and run your own code right in the browser.

Browser-based tools for Python are now abundant, thanks to Skulpt, a JavaScript 'port' of the Python interpreter. You can run basic Python code online at *www.codeskulptor.org* and *www.repl.it*

There are also some specialised Python-based online tools available. *www.morph.io* lets you write Python scrapers for extracting data from other websites; PythonAnywhere focuses on hosting Python code and Wakari lets you do scientific computing using Python.

Speaking of scientific computing, packages like NumPy allow one to perform extensive mathematical operations, SymPy allows for symbolic mathematics, Pandas is great for data manipulations, and matplotlib is brilliant for plotting data.

One of the greatest tools though is probably IPython, an enhanced Python shell that integrates well with the above to let you show plots embedded in your Python console. IPython notebook lets you create interactive documents that contain a mixture of rich text, plots and calculations.

There are even specialised versions of Python that come with these packages pre-installed, such as Enthought's Canopy, Continuum Analytics' Anaconda, and Python(x,y).

Speaking of specialised versions, it's important to note that the Python installers you get on python.org are just one implementation of the Python language. The three implementations mentioned above bundle tonnes of scientific tools with Python, then there is eGenix Python that's a single exe!

There are also complete re-implementations of Python. For instance JPython is a Python interpreter that runs on the Java VM, and as such it can access Java functionality. Similarly, you have IronPython that runs on .NET

Most interesting is PyPy, a Python implementation that was originally written in Python itself. Now it's a JIT compiler — the kind you find for JavaScript in modern browsers — for Python. In many cases it performs significantly faster than the standard Python implementation.